

Final Results — NeuroSleep

This is the single authoritative result document for the project. All other result tables in the repository are historical.

Executive Summary

Property	Value
Model	Improved Student — Full Fine-Tuning
Parameters	99,477
Dataset	Sleep-EDF Expanded (92 subjects)
Accuracy	87.7% ± 2.7%
Cohen's Kappa	0.763 ± 0.043
Macro F1	0.730 ± 0.037
Weighted F1	89.0% ± 2.1%
MGm	0.797 ± 0.040
Evaluation	10-fold subject-level CV, 3 seeds (42, 43, 44)

Per-Class Performance (Full Fine-Tuning, 30 Folds)

Stage	F1	Precision	Recall
Wake	0.964 ± 0.016	0.995 ± 0.003	0.936 ± 0.031
N1	0.445 ± 0.061	0.330 ± 0.065	0.712 ± 0.070
N2	0.768 ± 0.041	0.878 ± 0.043	0.687 ± 0.063
N3	0.681 ± 0.114	0.562 ± 0.135	0.896 ± 0.075
REM	0.771 ± 0.078	0.772 ± 0.077	0.785 ± 0.118

N1 Analysis

N1 is the hardest stage with F1=0.445. This is expected because:

- N1 epochs are brief (1-7 minutes) and transitional
- N1 is often confused with N2 and Wake
- N1 is only ~4.6% of the dataset

Model Strengths

- High Wake F1** (0.964) — Wake classification is excellent
- Strong N2 detection** (F1=0.768) — light sleep well-distinguished
- Balanced performance** — Macro F1 (0.730) indicates good stage balance

Adaptation Method Comparison (30 Folds)

Model	Trainable Params	Accuracy	κ	Macro F1	Weighted F1	MGm
Frozen Base	0 (0%)	87.1% ± 3.6%	0.738 ± 0.077	0.673 ± 0.074	87.1% ± 4.2%	0.668 ± 0.102
LoRA CNN+Head (r=8)	1,448 (1.43%)	83.6% ± 3.7%	0.693 ± 0.057	0.674 ± 0.045	85.5% ± 3.3%	0.744 ± 0.062
Full Fine-Tuning	99,477 (100%)	87.7% ± 2.7%	0.763 ± 0.043	0.730 ± 0.037	89.0% ± 2.1%	0.797 ± 0.040

Per-Stage F1 Comparison

Stage	Frozen	LoRA CNN+Head	Full FT
Wake	0.964 ± 0.020	0.945 ± 0.024	0.964 ± 0.016
N1	0.345 ± 0.082	0.357 ± 0.041	0.445 ± 0.061
N2	0.753 ± 0.055	0.721 ± 0.044	0.768 ± 0.041
N3	0.629 ± 0.114	0.668 ± 0.110	0.681 ± 0.114
REM	0.672 ± 0.157	0.675 ± 0.117	0.771 ± 0.078

Parameter Efficiency

- LoRA CNN+Head uses **68.7× fewer trainable parameters** (1,448 vs 99,477)
- LoRA CNN+Head retains **95.4% of full FT accuracy** (83.6% / 87.7%)
- LoRA CNN+Head retains **90.9% of full FT κ** (0.693 / 0.763)
- Frozen base already transfers well at **99.3% of full FT accuracy**

Key Findings

1. **Full fine-tuning achieves the strongest overall and stage-balanced performance**
2. **Frozen base transfers surprisingly well** — only 0.6% accuracy drop vs full FT
3. **LoRA provides parameter-efficient adaptation** but does not match full FT
4. **Full FT improves every stage** over frozen base, especially N1 (+0.10 F1) and REM (+0.10 F1)

Model Architecture

```
PSG Input (EEG + EOG + EMG, 4 channels)
    ↓
Multi-Resolution Stem
    ↓
Depthwise-Separable CNN
    ↓
Parametric Gabor FEB
    ↓
2-Layer GRU (300s context)
    ↓
5-Class Softmax
```

Property	Value
Parameters	99,477
Model Size	~400 KB (FP32)
Input Shape	[batch, 10, 4, 3000]
Output Shape	[batch, 10, 5]
Context Window	300 seconds (10 × 30s epochs)

Training Protocol (100-Subject)

Parameter	Value
Optimizer	AdamW
Learning Rate	3e-4
Weight Decay	1e-4
Epochs	20 (early stopping patience=5)
Batch Size	32
Class Weights	N1=2x, REM=2x
Gradient Clipping	max_norm=1.0
Scheduler	Cosine Annealing
Mixed Precision	True (CUDA)
Device	NVIDIA GTX 1650 (CUDA 12.4)
Seeds	42, 43, 44

Evaluation Protocol (100-Subject)

- **Method:** 10-fold subject-level cross-validation
- **Subjects:** 92 from Sleep-EDF Expanded (8 wake-only excluded)
- **Fold assignment:** Canonical (canonical_subject_folds_92subj.json)
- **Sequence:** length=10, stride=5, 30s epochs
- **All-position supervision** — every epoch in the window is supervised
- **Reproducibility:** 3 seeds × 10 folds = 30 folds per method

Files

100-Subject Benchmark (Authoritative)

File	Description
configs/full_100_subject.yaml	100-subject configuration
artifacts/final/student_full_finetuned.pt	Checkpoint
results/100_subject_adaptation/final/aggregate_metrics.json	3-seed aggregate
results/100_subject_adaptation/final/overall_comparison.csv	Overall metrics
results/100_subject_adaptation/final/per_class_comparison.csv	Per-class metrics
results/100_subject_adaptation/final/FINAL_ADAPTATION_RESULTS.md	Full report

15-Subject Development (Historical)

File	Description
configs/final.yaml	Development configuration
results/final/final_metrics.json	Development results

Reproduction

```
# Run 100-subject adaptation benchmark (all 3 modes, 1 seed)
python scripts/train_adaptation.py --mode frozen --seed 42 --device cuda
python scripts/train_adaptation.py --mode lora --targets enc.0.pw,enc.1.pw,head --rank 8 --alpha 16 --seed 42 --device cuda
python scripts/train_adaptation.py --mode full_finetune --seed 42 --device cuda

# Aggregate all 3 seeds
python scripts/aggregate_adaptation_results.py

# Verify protocol consistency
python scripts/protocol_fingerprint.py --seed 42 --mode full_finetune --save-reference
python scripts/protocol_fingerprint.py --seed 43 --mode full_finetune
```

Citation

```
@project{neurosleep_2026,
  title={NeuroSleep: Neuromorphic Sleep Stage Scoring},
  author={Kaushik, P. and Vora, S. and Bhatt, S. and Khan, S. and Lone, A.J.},
  year={2026},
  institution={VIT Bhopal University}
}
```

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